

Homework #2:

Due date: Oct. 9th 2025

1. Normal modes of a square lattice:

Consider a square lattice of atoms of mass M , in which the nearest neighbors are coupled to each other by springs with constants K . Derive the equation of motion for the mode in which atomic displacements are normal to the crystal plane and find the dispersion of this mode. Analyze the behavior of this mode in the continuum limit.

2. Localized phonons on an impurity:

Consider a linear chain of identical atoms of mass m each equidistant by a .

- Limiting the problem to interactions between nearest neighbors characterized by the force constant β , give the dispersion relation of longitudinal phonons and derive the expression of their group velocity. Express the results as a function of " a " and of the velocity of sound, v_s . What is maximum frequency of atomic vibration, ω_M , when $v_s = 5000$ m/s?
- An atom in the middle of the chain is replaced by the atom of an impurity with a mass m' (where $m' < m$). Write the displacement equations, u_0 , for this impurity atom and the displacement " u_n " for an atom located at a distance " na ." Assume that the force constant β between nearest neighbors is the same regardless of the atom considered.
- The impurity atom being lighter than the others has an oscillation frequency $\omega_{m'}$ larger than ω_M and it drags the other neighbor atoms by its motion. We try to describe its effect with the expression of a damped wave of form: $u_n = u_0 e^{-a|x|} e^{i(\omega't - kx)}$. What is the value of the frequency ω'_M , which is compatible with the equations of motion when the wave vector is near the boundary of the first BZ? Verify that when $m = m'$, the solution matches that obtained in (a).
- After having numerically evaluated ω'_M and a when $m' = m/2$, sketch the instantaneous position of atoms as a function of x .

3. Density of states and specific heat of a monoatomic 1D lattice:

Consider a row of N identical atoms of mass m and equidistant by a .

- Limiting the problem to interactions between nearest neighbors with a force constant β , find the expression for the dispersion relation of acoustic longitudinal vibrations propagating along this row. What is the maximum frequency ω_M of atomic vibrations, written in terms of v_s and a ? Compare the value obtained with that of ω_D deduced from the Debye model.
- Find the corresponding expression for the density of states $g(\omega)$.
- Write the expressions for internal energy U and for the heat capacity C_v . Find the limits as these functions approach high temperatures ($\hbar\omega_M \ll k_B T$).
- What is the evolution of C_v at low temperature? Compare the result obtained here with that deduced from the Debye model